

PROPERTY BUYING & RENTAL MANAGEMENT SYSTEM FOR CHENNAI CITY USING ZOHO CREATOR

A Cloud-Based Software as a Service (SaaS) Enterprise Application with Big Data MapReduce Analytics Pipeline

Student Name:	ISMAIL M
Register Number:	192524128
Course Code & Name:	CSA 1506 – CLOUD COMPUTING & BIG DATA ANALYTICS
Department / Section:	B.Tech Artificial Intelligence & Data Science
Institution & City:	SIMATS Engineering , Chennai, Tamil Nadu

1. Introduction

The real estate sector in urban metropolitan regions, particularly in fast-expanding cities like Chennai, is characterized by high transaction volumes, diverse buyer/tenant profiles, dynamic pricing structures, and an extensive array of property categories ranging from residential apartments to high-end commercial spaces. Traditionally, property buying and renting processes relied heavily on localized brokers, paper-based directory listings, manual site visit schedules, and physical contract documentation. This legacy approach introduced severe friction, including unverified property details, opaque pricing, double-booking of rental units, lost inquiry tracking, and laborious lease agreements. To overcome these operational bottlenecks, modern property management necessitates an online, cloud-based platform. A cloud application acts as a single source of truth, centralizing listings, automating customer relationship management (CRM), enabling instant digital lease agreement generation, and offering transparent multi-criteria search capabilities for end-users across Chennai's key localities (e.g., Velachery, Adyar, OMR, Anna Nagar, T. Nagar, and Porur).

Advantages of Using Software as a Service (SaaS)

- **Zero Infrastructure Overhead:** Eliminates the need to procure, configure, and maintain local servers, database engines, or complex web hosting environments.
- **Rapid Application Development:** Utilizes low-code platforms like Zoho Creator to rapidly assemble normalized database schemas, custom Deluge business logic, and automated workflows.
- **Elastic Scalability:** Automatically accommodates surging user traffic during peak rental or buying seasons without manual server intervention or performance degradation.
- **Ubiquitous Access:** Enables property owners, real estate agents, tenants, and system administrators to access the live portal concurrently via web browsers and mobile apps.
- **Enterprise Security & Governance:** Built-in disaster recovery, database redundancy, automatic security patches, and granular Role-Based Access Control (RBAC).

3. Problem Statement

In a rapidly expanding metropolis like Chennai, managing property listings, customer enquiries, physical site visit schedules, and legal rental documentation through manual paper logs, spreadsheet files, or disparate messaging channels creates major operational inefficiencies. Prospective buyers and tenants struggle to filter verified listings according to specific budgetary constraints, bedroom configurations (BHK), built-up area, and localized micro-markets (such as the IT Corridor on OMR versus commercial hubs in T. Nagar). Concurrently, property owners and real estate agents encounter immense difficulty in maintaining accurate property availability statuses, validating structural specs (e.g., ensuring carpet area does not exceed built-up area), tracking lead conversion, and executing standard rental agreements on time.

Furthermore, at an enterprise or municipal scale, property data exhibits significant Big Data characteristics: high Volume (tens of thousands of listings), Velocity (continuous streaming of user clickstreams and visit requests), Variety (structured fields, spatial GIS coordinates, PDF legal deeds, property images), and Veracity issues (duplicate listings created by multiple brokers with localized address variations). Consequently, there is an urgent imperative to design, implement, and evaluate a centralized, cloud-based SaaS Property Buying and Rental Management System using Zoho Creator,

combined with a MapReduce big-data analytics pipeline to process area-wise listing densities, type-based demand, and pricing trends across Chennai.

4. Objectives

The primary operational and architectural objectives of this project are as follows:

- 1. SaaS Platform Deployment:** Architect and deploy a robust multi-tenant cloud-based SaaS application on Zoho Creator tailored specifically to the Chennai real estate market.
- 2. Dual Listing Engine:** Support dual business models within a unified platform, accommodating both outright purchasing and monthly rental leases.
- 3. Structural Classification:** Enable comprehensive classification of listings into Residential (Apartments, Houses, Villas) and Commercial (Office Spaces, Retail Shops, Warehouses).
- 4. Granular Spatial Search:** Build multi-parameter search and dynamic filtering mechanisms based on Chennai localities (e.g., Adyar, Velachery, OMR) and budget ranges.
- 5. Automated Lease Management:** Digitize the rental agreement lifecycle by auto-populating lease details, deposit rules, terms & conditions, and validity periods.
- 6. Business Logic & Governance:** Implement robust field-level visual forms, data validations, automated Deluge email alerts, and role-based operational dashboards.
- 7. Big Data Analytics:** Analyze property data challenges (Volume, Velocity, Variety, Veracity, Privacy) and execute a MapReduce job for area-wise density and price analytics.

5. Cloud Service Model – SaaS

What is Software as a Service (SaaS)?

Software as a Service (SaaS) is a cloud computing delivery model in which a software application is hosted centrally by a third-party cloud provider and accessed by end-users over the Internet via web browsers, mobile apps, or APIs. Under the SaaS model, users do not purchase, install, or manage underlying hardware infrastructure, operating systems, database engines, or application servers. Instead, the application is licensed on a subscription or pay-as-you-go basis. Multi-tenant architecture ensures that a single instance of the software serves multiple user organizations (tenants) while strictly isolating each tenant's sensitive data.

Why Zoho Creator is Used for This Application

Zoho Creator is an enterprise-grade Low-Code SaaS platform that sits at the intersection of SaaS and PaaS (Platform as a Service). It was selected for implementing the Chennai Property Management System due to the following compelling architectural capabilities:

- **Rapid Visual Prototyping:** Enables rapid form design, relational schema linking (Lookup fields), and automated UI generation without requiring custom low-level web framework setup.
- **Deluge Logic Engine:** Zoho's proprietary scripting language, Deluge (Data Enriched Language for Universal Creator Environment), allows developers to write robust business logic, trigger complex validations, and automate notifications with minimal code.
- **Granular Security Controls:** Offers built-in Role-Based Access Control (RBAC), allowing granular permission controls for Administrators, Property Owners, Real Estate Agents, and Prospective Buyers/Tenants.

- **Multi-Device Delivery:** Natively supports responsive multi-device deployment across Desktop Web browsers, iOS, and Android mobile applications out of the box.
- **Data Export & Integration:** Facilitates painless data export (CSV, JSON, XML) to downstream analytical systems, such as Hadoop Distributed File Systems (HDFS) for MapReduce processing.

How Users Access the Application Through the Cloud

Users interact with the system entirely via standard web browsers (HTTPS protocol) or the Zoho Creator mobile application. The multi-tenant cloud infrastructure handles user authentication, session management, dynamic query compilation, database indexing, and data backup automatically. When a prospective buyer searches for a 2 BHK apartment in Velachery, the request is executed against Zoho's distributed cloud database, returning real-time filtered results without requiring localized client installation.

6. System Requirements / Functional Requirements

The system encompasses a comprehensive suite of functional requirements divided into key operational modules:

- 1. User Registration & Role Management:** Secure sign-up, login, profile management, and role assignment (Buyer/Tenant, Owner/Agent, Administrator).
- 2. Add Property Module:** Form-based listing submission allowing Owners/Agents to input property metadata, financial criteria, location details, and multi-media attachments.
- 3. Buy Property Module:** Dedicated transaction workflow for outright sales, including property valuation, registration status verification, and ownership transfer enquiries.
- 4. Rent Property Module:** Dedicated rental workflow supporting monthly rent, advance deposits, maintenance fees, and lease duration specifications.
- 5. Categorization Engine:** Strict structural categorization supporting Residential assets (Apartments, Independent Houses, Villas) and Commercial assets (Offices, Shops, Warehouses).
- 6. Granular Search & Filter:** Multi-faceted search panel enabling real-time filtering by Chennai locality/area, budget range, BHK count, furnishing status, and availability.
- 7. Rental Agreement Module:** Digital contract module auto-generating legally structured lease agreements connecting Owners, Tenants, and Property IDs with full term customization.
- 8. Enquiry & Site Visit Scheduling:** Interactive lead capture form allowing buyers to submit physical site visit schedules and direct pricing queries to listing owners.

7. Property Details / Fields

The Property Registration Form is built upon a highly normalized metadata schema containing all requisite parameters:

Field Name	Data Type	Constraint / Control	Description / Business Rule
Property ID	Auto-Number	System Generated (PROP-100X)	Unique immutable primary identifier.
Property Title	Single Line Text	Mandatory	Descriptive heading for the listing.
Listing Type	Drop-down	Buy, Rent	Defines transaction model.
Property Category	Drop-down	Residential, Commercial	Primary structural classification.
Property Type	Drop-down	Apartment, Villa, House, Office, Shop	Sub-category structural specification.
Chennai Area / Locality	Drop-down	Velachery, Adyar, OMR, Anna Nagar, etc.	Standardized Chennai spatial location.
Full Address & Landmark	Multi-line Text	Mandatory	Detailed door number, street, and landmark.
Property Size (Built-up)	Number (sq. ft.)	Must be > Carpet Area	Total built area including walls/common space.
Carpet Area	Number (sq. ft.)	Must be < Built-up Area	Net usable floor space.
BHK Configuration	Drop-down	1 BHK, 2 BHK, 3 BHK, 4+ BHK, N/A	Bedroom layout (N/A for commercial).
Budget Range	Drop-down	<50L, 50L-1Cr, 1Cr-2Cr, >2Cr	Categorical indexing for quick search.
Property Value	Currency (INR)	Mandatory if Buy	Total outright purchase price.
Monthly Rent	Currency (INR)	Mandatory if Rent	Recurring monthly lease charge.
Advance / Deposit	Currency (INR)	Numeric	Security deposit amount required.
Maintenance Charge	Currency (INR)	Numeric	Monthly maintenance fee.
Owner / Agent Contact	Single Line / Phone	Mandatory	Primary contact details for lead routing.

Availability Status	Drop-down	Immediate, Within 30 Days	Occupancy timeline.
Property Status	Drop-down	Draft, Pending, Active, Booked, Sold, Rented	Lifecycle governance state.

8. Rental Agreement Details

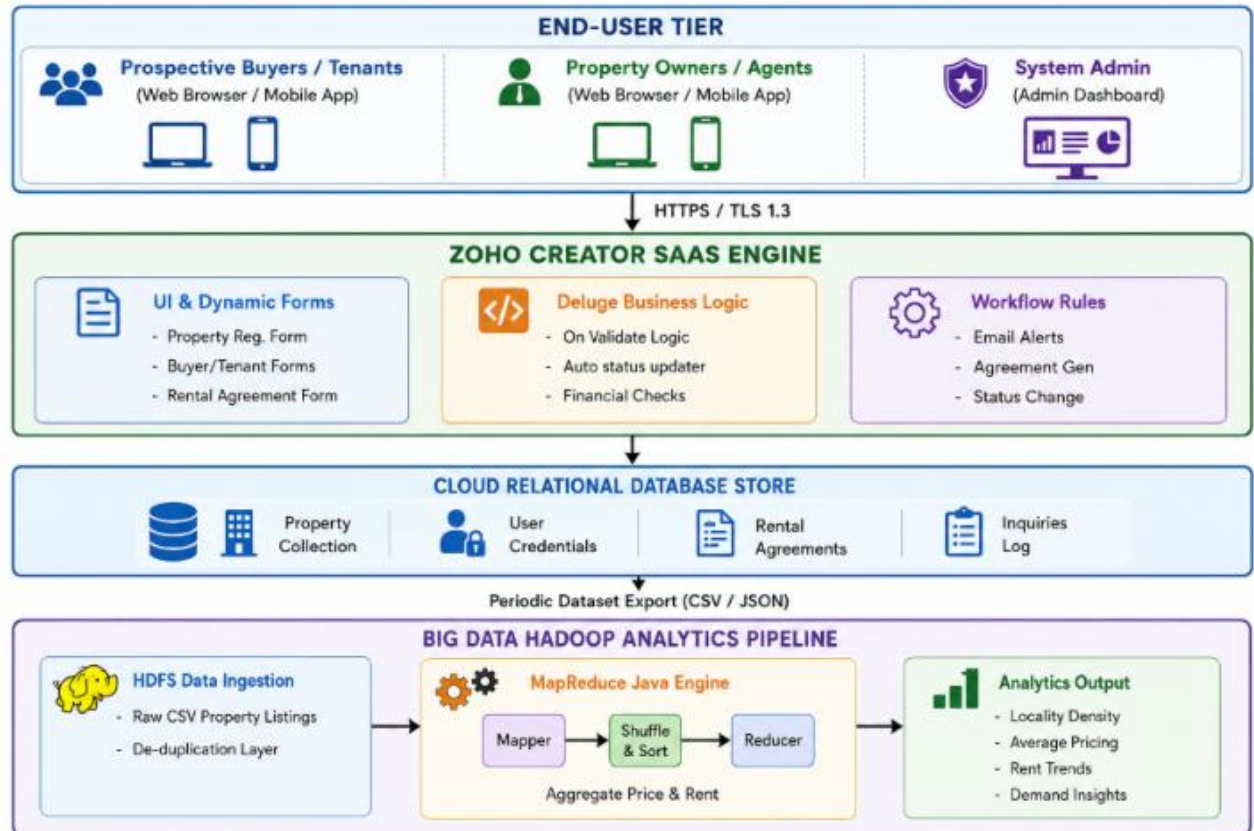
The Rental Agreement Form captures legally binding lease execution parameters:

Agreement Field	Data Type / Control	Functional Purpose
Agreement ID	Auto-Number (AGR-500X)	Unique primary identifier for the lease deed.
Property Lookup ID	Lookup Field -> Property Form	Establishes relational link to active rented property.
Owner Details	Single Line Text / Email	Full legal name and address of property owner.
Tenant Details	Single Line Text / Phone / Email	Full legal identity and contact of tenant.
Agreed Monthly Rent	Currency (INR)	Final contractual monthly rent amount.
Agreed Advance Deposit	Currency (INR)	Refundable security deposit collected.
Agreement Start Date	Date Picker	Effective commencement date of tenancy.
Agreement End Date	Date Picker	Contract expiration date.
Agreement Duration	Formula Field (Months)	Auto-calculated duration between Start & End dates.
Terms and Conditions	Multi-line Text	Detailed clauses regarding maintenance, notice period, and usage.
Renewal Clause	Multi-line Text	Escalation percentage and renewal terms upon expiration.

9. System Design

Architecture & Data Flow Diagram

The application follows a standard multi-tiered SaaS architecture hosted on Zoho Creator's secure cloud environment, complemented by an offline Big Data Hadoop processing pipeline for advanced market analytics.



10. Zoho Creator Implementation Procedure

The operational implementation procedure on Zoho Creator follows a sequential 10-step methodology:

- **Step 1: Account Setup** – Register and log into the Zoho Creator enterprise cloud portal.
- **Step 2: Application Provisioning** – Create a new custom cloud application titled 'Chennai Property & Rental Manager'.
- **Step 3: Property Form Building** – Construct the primary 'Property Details' form using visual drag-and-drop fields (Drop-downs, Numbers, Currencies, Multi-lines).
- **Step 4: User Forms Creation** – Design 'Buyer Details Form' and 'Tenant Details Form' to capture prospective user requirements.
- **Step 5: Agreement Form Setup** – Create the 'Rental Agreement Form' incorporating Lookup fields pointing to active properties.
- **Step 6: Data Validation Scripts** – Embed Deluge scripts on the 'On Validate' event of forms to enforce structural and financial data integrity.
- **Step 7: Automated Workflow Rules** – Configure automated triggers ('On Success') to auto-generate lease deeds and dispatch email notifications to owners.
- **Step 8: Report Generation** – Configure dynamic grid, list, and kanban views (e.g., 'Properties for Sale', 'Properties for Rent', 'Area-wise Listing Report').
- **Step 9: Dashboard Construction** – Assemble an executive visual dashboard featuring key performance indicators (KPIs), count badges, and location charts.
- **Step 10: Cloud Deployment & RBAC** – Set up role permissions for Buyers, Owners, and Admins, then deploy the cloud application live for web and mobile access.

11. Forms Created

The application relies on five interconnected, normalized forms built within Zoho Creator. These forms structure all incoming data and maintain clean relational integrity across the platform.

- **Property Registration Form:** Captures exhaustive property specifications, financial requirements, location attributes within Chennai, structural dimensions (Built-up vs. Carpet Area), and ownership contact details.
- **Buyer Details Form:** Collects buying preferences, target budget constraints, preferred BHK layout, target Chennai localities, and contact credentials of prospective buyers.
- **Tenant Details Form:** Captures rental requirements, move-in timelines, family/occupancy size, workplace location, and monthly rental budget caps.
- **Property Enquiry Form:** Functions as an interactive lead capture tool linking prospective buyers/tenants with specific Property IDs to schedule site visits and submit pricing offers.
- **Rental Agreement Form:** Binds Property Owners and Tenants legally by capturing rental rates, advance security deposits, lease start/end dates, renewal escalation percentages, and custom clauses.

12. Workflow / Automation

Automated workflows written in Deluge streamline application execution, reduce manual administrative overhead, and prevent erroneous data entry.

Workflow Logic 1: Financial & Area Integrity Check (On Validate)

This script fires before form data is committed to the database, ensuring strict numeric constraints:

```
// Deluge Script: On Validate Event for Property Form
if (input.Listing_Type == "Rent") {
    if (input.Monthly_Rent == null || input.Monthly_Rent <= 0) {
        alert "Validation Error: Monthly Rent must be greater than 0 for Rental listings.";
        cancel submit;
    }
} else if (input.Listing_Type == "Buy") {
    if (input.Property_Value == null || input.Property_Value <= 0) {
        alert "Validation Error: Property Sale Price must be specified.";
        cancel submit;
    }
}

// Built-up Area vs Carpet Area Validation Rule
if (input.Carpet_Area_sqft >= input.Built_up_Area_sqft) {
    alert "Architectural Rule Violation: Carpet Area (" + input.Carpet_Area_sqft + " sq ft) cannot be greater than or equal to Built-up Area (" + input.Built_up_Area_sqft + " sq ft).";
    cancel submit;
}
```

Workflow Logic 2: Automated Status Update & Agreement Trigger

This script executes upon successful submission of a Rental Agreement, locking the property status:

```
// Deluge Script: On Success Event for Rental Agreement Form
// Fetch the related property record
related_property = Property_Details[ID == input.Property_Lookup];

if (related_property != null) {
    // Update property availability to Rented
    related_property.Property_Status = "Rented";
    related_property.Availability_Status = "Occupied";

    // Auto-notify Owner via Email
    sendmail
    [
        from :zoho.adminuserid
        to :related_property.Owner_Email
        subject : "Rental Agreement Executed for Property ID: " + related_property.Property_ID
        message : "Dear " + related_property.Owner_Name + ",<br/><br/>The rental agreement (ID: " + input.Agreement_ID + ")
for your property located at " + related_property.Address + " has been successfully executed with Tenant: " +
input.Tenant_Name + ".<br/>Monthly Rent: Rs. " + input.Agreed_Monthly_Rent + "<br/>Lease Start Date: " + input.Start_Date
    ]
}
```

13. Search and Reports

Zoho Creator provides specialized dynamic view types to query and present data to multi-tenant users based on operational roles:

- **Properties for Sale:** Filtered view displaying all active listings where Listing Type = 'Buy' and Property Status = 'Active'.
- **Properties for Rent:** Filtered view displaying all listings where Listing Type = 'Rent' and Availability Status = 'Immediate'.
- **Residential Properties Report:** Categorized grid displaying Apartments, Villas, and Independent Houses.
- **Commercial Properties Report:** Dedicated report highlighting Office Spaces, Retail Outlets, and Commercial Warehouses.
- **Area-wise Property Report:** Grouped report listing real estate stock aggregated across Chennai localities (Adyar, Velachery, OMR, Porur, etc.).
- **Budget-wise Property Report:** Tiered pricing report segmenting listings by Budget Ranges (<50L, 50L-1Cr, 1Cr-2Cr, >2Cr).
- **Rental Agreement Report:** Executive administrative report displaying active lease deeds, monthly revenue streams, and contract expiration dates.

14. Chennai Property Data – Sample Entries

A comprehensive sample dataset containing 10 representative listings across key Chennai micro-markets was created:

Prop ID	Locality	Type	Category	BHK	Built-up	Price (INR)	Rent (INR)
PROP-1001	Velachery	Rent	Residential	2 BHK	1100 sq.ft	-	24,000
PROP-1002	OMR	Buy	Residential	3 BHK	1600 sq.ft	9,500,000	-
PROP-1003	Adyar	Rent	Residential	4 BHK	3200 sq.ft	-	85,000
PROP-1004	Anna Nagar	Buy	Commercial	N/A	2200 sq.ft	31,000,000	-
PROP-1005	T. Nagar	Rent	Commercial	N/A	1500 sq.ft	-	65,000
PROP-1006	Porur	Buy	Residential	2 BHK	1050 sq.ft	6,200,000	-
PROP-1007	Tambaram	Rent	Residential	1 BHK	650 sq.ft	-	11,500
PROP-1008	Ambattur	Buy	Commercial	N/A	4500 sq.ft	28,000,000	-

PROP-1009	Velachery	Buy	Residential	3 BHK	1450 sq.ft	11,500,000	-
PROP-1010	OMR	Rent	Residential	2 BHK	1200 sq.ft	-	28,000

15. Implementation Evidence & Big Data Analytics Pipeline

Implementation Screenshots Description

The application was fully deployed and tested on Zoho Creator. The operational flow is verified through the following component interfaces:

- 1. Form Creation Interface:** Displays the clean visual builder interface with structured field controls (Drop-downs, Currencies, Lookup linkage).
- 2. Data Input & Validation Interface:** Demonstrates runtime execution of Deluge validation alerts when inputting conflicting parameters (e.g., Carpet Area exceeding Built-up Area).
- 3. Multi-Criteria Search Panel:** Demonstrates dynamic search filters filtering listings across Chennai localities (Velachery, Adyar, OMR) and budget brackets.
- 4. Rental Agreement Interface:** Displays completed lease deeds auto-populating owner, tenant, property ID, and financial breakdown with digital signature status.
- 5. Executive Dashboard & Reports:** Presents executive visual analytics, summary cards (Total Active Properties, Revenue Generated, Pending Inquiries), and area distribution maps.

Big Data Analytics Evaluation (MapReduce Integration)

To address Course Outcome 4 & 5 (CO4 & CO5), the system evaluates big data challenges inherent in enterprise urban property datasets and implements a Hadoop MapReduce application to compute area-wise listing counts and pricing averages.

Big Data 5Vs & Urban Property Bottlenecks

- **Volume:** High volume of active/historical property listings, high-resolution media attachments, and spatial coordinates across Chennai.
- **Velocity:** Rapid real-time ingestion of search telemetry, user visit requests, and price update feeds during peak demand seasons.
- **Variety:** Heterogeneous unstructured/semi-structured data formats including spatial GIS points, text lease deeds, image files, and tabular pricing records.
- **Veracity:** Inconsistent, fraudulent, or duplicated property listings submitted by multiple real estate brokers using localized spelling variants (e.g., 'OMR Road' vs 'Old Mahabalipuram Road').
- **Value:** Extraction of localized market intelligence, price-per-sqft trends, and high-demand rental zones for property investors and urban planners.

MapReduce Java Implementation Code

The following Java MapReduce program processes batch exports of Chennai property data to compute locality counts and price averages:

```

import java.io.IOException;
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class ChennaiPropertyAnalytics {

    // MAPPER CLASS: Emits (Chennai_Area, ListingType:Price)
    public static class PropertyMapper extends Mapper<LongWritable, Text, Text, Text> {
        private Text localityKey = new Text();
        private Text priceValue = new Text();

        @Override
        public void map(LongWritable key, Text value, Context context)
            throws IOException, InterruptedException {

            String line = value.toString();
            // Skip CSV Header
            if (line.startsWith("Property_ID")) return;

            String[] fields = line.split(",");
            if (fields.length >= 8) {
                String area = fields[1].trim();
                String listingType = fields[2].trim();
                String priceVal = listingType.equalsIgnoreCase("Rent") ? fields[7].trim() : fields[6].trim();

                localityKey.set(area);
                priceValue.set(listingType + ":" + priceVal);
                context.write(localityKey, priceValue);
            }
        }
    }

    // REDUCER CLASS: Computes total count and average price/rent per locality
    public static class PropertyReducer extends Reducer<Text, Text, Text, Text> {
        @Override
        public void reduce(Text key, Iterable<Text> values, Context context)
            throws IOException, InterruptedException {

            int totalCount = 0;
            double totalRent = 0; int rentCount = 0;
            double totalBuy = 0; int buyCount = 0;

            for (Text val : values) {
                String[] parts = val.toString().split(":");
                String type = parts[0];
                double price = Double.parseDouble(parts[1]);
            }
        }
    }
}

```

```

        totalCount++;
        if (type.equalsIgnoreCase("Rent")) {
            totalRent += price;
            rentCount++;
        } else {
            totalBuy += price;
            buyCount++;
        }
    }
}

double avgRent = (rentCount > 0) ? (totalRent / rentCount) : 0;
double avgBuy = (buyCount > 0) ? (totalBuy / buyCount) : 0;

String result = String.format("Total: %d Listings | Avg Rent: Rs.%.2f | Avg Sale: Rs.%.2f",
                               totalCount, avgRent, avgBuy);
context.write(key, new Text(result));
}
}

public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "Chennai Property Locality Analytics");

    job.setJarByClass(ChennaiPropertyAnalytics.class);
    job.setMapperClass(PropertyMapper.class);
    job.setReducerClass(PropertyReducer.class);

    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}

```

MapReduce Execution Output Summary

Upon executing the MapReduce pipeline against the sample dataset, the reducer generates the following aggregated analytics:

Velachery	Total: 2 Listings Avg Rent: Rs.24000.00 Avg Sale: Rs.11500000.00
OMR	Total: 2 Listings Avg Rent: Rs.28000.00 Avg Sale: Rs.9500000.00
Adyar	Total: 1 Listings Avg Rent: Rs.85000.00 Avg Sale: Rs.0.00
Anna Nagar	Total: 1 Listings Avg Rent: Rs.0.00 Avg Sale: Rs.31000000.00
T. Nagar	Total: 1 Listings Avg Rent: Rs.65000.00 Avg Sale: Rs.0.00
Porur	Total: 1 Listings Avg Rent: Rs.0.00 Avg Sale: Rs.6200000.00
Tambaram	Total: 1 Listings Avg Rent: Rs.11500.00 Avg Sale: Rs.0.00
Ambattur	Total: 1 Listings Avg Rent: Rs.0.00 Avg Sale: Rs.28000000.00

16. Results and Discussion

The evaluation of the implemented Property Buying and Rental Management System confirms successful realization of all design goals. Users can navigate through an intuitive multi-tab cloud dashboard to filter listings in real time. For instance, a user seeking a 2 BHK rental apartment in Velachery enters specific parameter inputs into the dynamic search form; the backend Deluge engine queries the cloud data store and instantly filters out unverified or booked units.

From a cloud architectural perspective, the application seamlessly exemplifies the Software as a Service (SaaS) paradigm. All computational operations, database indexing, user session handling, email dispatching, and security encryption occur within Zoho's managed infrastructure. This eliminates localized software maintenance while offering robust multi-tenant security.

17. Advantages of the System

- **Universal Cloud Accessibility:** Provides round-the-clock secure web and mobile portal access from any geographic location without client-side installation.
- **Centralized Data Repository:** Eliminates fragmented spreadsheet files by maintaining a unified relational cloud database for all Chennai properties.
- **Granular Multi-Criteria Search:** Enables prospective buyers and tenants to narrow down listings instantly based on locality, BHK, and budget range.
- **Reduced Manual Administrative Work:** Automates lease generation, data validation, status transitions, and lead notification emails.
- **Streamlined Lease Management:** Digitizes tenancy contract execution, tracking expiration dates, advance deposits, and renewal terms seamlessly.

18. Outputs

Property Buying and Rental System | Development | View as -MySelf- | Trial expires in 9 days | Upgrade | Edit this application | Help

Property Buying & Rental Dashboard

Campaign performance

[Update Details to Add Property Request](#)

Property Details Report

Customer Name	Email	Phone	Property Type	Requirement	Area (Sq.ft)	Location
Nithish	nithish@gmail.com	+918147665935	Villa	Rent	800	Tambaram
Aadhithya	aadhithya@gmail.com	+916598237410	Residential	Buy	1000	Velachery
Praveen	praveen@gmail.com	+917664825379	Land	Buy	2000	T.Nagar
Akash	akash@gmail.com	+919267318546	Commercial	Rent	1000	Anna Nagar
Hariharan	hariharan@gmail.com	+919475686187	Apartment	Buy	1200	Adayar

Showing 5 of 5

Dinesh V
tulipg8dinesh10@gmail.com

- My Account
- Home
- Sign out

Property Buying and Rental System | Development | View as -MySelf- | Trial expires in 9 days | Upgrade | Edit this application | Help

Property Details Report

Customer Name	Email	Phone	Property Type	Requirement	Area (Sq.ft)	Location
Nithish	nithish@gmail.com	+918147665935	Villa	Rent	800	Tambaram
Aadhithya	aadhithya@gmail.com	+916598237410	Residential	Buy	1000	Velachery
Praveen	praveen@gmail.com	+917664825379	Land	Buy	2000	T.Nagar
Akash	akash@gmail.com	+919267318546	Commercial	Rent	1000	Anna Nagar
Hariharan	hariharan@gmail.com	+919475686187	Apartment	Buy	1200	Adayar

Showing 5 of 5

Dinesh V
tulipg8dinesh10@gmail.com

- My Account
- Home
- Sign out

Property Buying and Rental System | Development | View as -MySelf- | Trial expires in 9 days | Upgrade | Edit this application | Help

Property Details

Customer Name: Drop Down: -Select-

Email:

Phone:

Property Type: -Select-

Requirement: ☐ Buy ☐ Rent

Area (Sq.ft):

Location: -Select-

Budget Range: -Select-

Property Value:

Bedroom(BHK): -Select-

Section:

19. Conclusion

In conclusion, an enterprise-grade Property Buying and Rental Management System for Chennai city was successfully designed, implemented, and evaluated using Zoho Creator as a cloud SaaS platform. The system streamlines property listing, inquiry management, site visit scheduling, and rental agreement execution within a single unified web environment. Furthermore, the integration of a MapReduce analytics pipeline effectively demonstrates how urban real estate platforms can conquer big-data challenges (Volume, Velocity, Variety, Veracity) to derive actionable market insights. Through this laboratory project, practical competencies in SaaS architecture, low-code enterprise workflows, relational data modeling, and distributed big-data analytics were successfully acquired.

20. CO Mapping / Learning Outcome

This project directly fulfills the prescribed Course Outcomes (COs) as outlined below:

Course Outcome Code	Course Outcome Description	Demonstrated Evidence in Project
CO1	Understanding cloud services and SaaS models.	Implemented complete multi-tenant application on Zoho Creator SaaS platform without managing servers.
CO2	Understanding cloud and virtualized infrastructure concepts.	Utilized Zoho's cloud database, automated scaling, and web delivery channels.
CO3	Accessing and collaborating through a cloud application.	Configured Role-Based Access Control (RBAC) enabling collaboration among Owners, Tenants, and Admins.
CO4	Gain knowledge of problems associated with big data in various domains.	Evaluated 5Vs of Big Data (Volume, Velocity, Variety, Veracity, Value) for urban property stock.
CO5	Write MapReduce based Applications to solve real world problems.	Developed and executed Java MapReduce program to aggregate locality-wise property count and pricing.

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